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Introduction

Anomaly Detection Systems (ADS) play a crucial role in modern cybersecurity by identifying unusual network traffic patterns that may indicate cyberattacks. Traditional rule-based ADS rely on predefined attack signatures, making them effective only against known threats. However, these systems struggle with zero-day attacks—new or evolving cyber threats that do not match any existing signatures.

Machine Learning (ML)-based ADS provide a more adaptive approach by learning from network behavior and identifying anomalies in real time. This study explores various ML models—Random Forest, XGBoost, LightGBM, and Extra Trees—to determine the most effective algorithm for anomaly detection. Using the ADS2018 dataset, we assess the capability of these models in distinguishing normal network traffic from potentially malicious activity.

Methods

Data Preprocessing – Cleaned the ADS2018 dataset, handled missing values, encoded categorical data, and normalized features.

Feature Selection – Used Recursive Feature Elimination (RFE) to retain key network traffic attributes, reducing redundancy.

Model Training – Evaluated **Random Forest, XGBoost, LightGBM, and Extra Trees** for anomaly detection.

Performance Analysis – Compared models using **accuracy, precision, recall, and F1-score** to assess effectiveness.

Data Analysis

Analyzing network traffic data is essential to improving the efficiency of anomaly detection models. Key aspects of the analysis include:

- **Feature Importance Analysis:** Identified which network attributes (e.g., IP addresses, packet size, connection duration, and protocol types) are most influential in detecting anomalies.
- **Class Imbalance Handling:**

In real-world network traffic, malicious attacks occur less frequently than normal activity.

Resampling techniques (such as Synthetic Minority Over-sampling Technique - SMOTE) were applied to balance the dataset and prevent biased predictions.

- **Visualization of Network Behavior:** Created data visualizations (e.g., packet distribution graphs) to illustrate normal vs. anomalous traffic.

Findings and Results

Anomaly Detection Accuracy: LightGBM demonstrated the highest test accuracy at 91.04%, making it the most effective model for anomaly detection. Feature selection significantly improved model efficiency by reducing computational complexity while maintaining high performance. ML-based anomaly detection provided a dynamic approach to identifying threats, unlike traditional signature-based systems that rely on predefined attack patterns.

Comparison of Models: LightGBM emerged as the best-performing model, balancing accuracy and computational efficiency. Random Forest and XGBoost showed competitive performance, but LightGBM outperformed them in recall and F1-score. Extra Trees provided similar results to Random Forest, while AdaBoost performed the worst, proving ineffective for this task.

Dashboard Visualization: Model performance insights and real-time anomaly detection results were visualized using dashboards, enabling security analysts to quickly identify and respond to threats. These dashboards provide an intuitive interface for monitoring network traffic patterns and assessing potential cyber threats efficiently.

Conclusions

The Anomaly Detection System demonstrated that using machine learning algorithms like evaluated models, LightGBM demonstrated the highest accuracy of 91.04%, making it the most suitable for detecting network anomalies efficiently. The implementation of feature selection and data preprocessing significantly contributed to improved model performance by optimizing computational efficiency. Unlike traditional signature-based detection systems, ML-based approaches offer adaptability in identifying emerging threats dynamically. Future research will focus on real-time deployment of these models and the integration of deep learning techniques to further enhance the accuracy and scalability of anomaly detection in cybersecurity applications.

References

1. K. Scarfone and P. Mell, *Guide to Intrusion Detection and Prevention Systems (IDPS)*, NIST, 2007.
2. M. Chandrashekhar and N. Sahasrabudhe, *A Survey on Anomaly-Based Intrusion Detection Systems*, IJCA, 2017.
3. T. Chen and C. Guestrin, *XGBoost: A Scalable Tree Boosting System*, KDD, 2016.
4. G. Ke et al., *LightGBM: A Highly Efficient Gradient Boosting Decision Tree*, NIPS, 2017.
5. Y. LeCun, Y. Bengio, and G. Hinton, *Deep Learning*, Nature, 2015.
6. H. He and E. A. Garcia, *Learning from Imbalanced Data*, IEEE Transactions, 2009.